1) Write a java Application to implement interfaces.

:-

```
interface A
{
	void display();
}
public class Demo implements A
{
	int x=10;
	public void display()
	{
		System.out.println("I am in interface");
	}
	public void show()
	{
		System.out.println("x="+x);
	}

	public static void main(String[] args) {
		Demo d1=new Demo();
		d1.display();
		d1.show();

	}

}
```

Output:

```
I am in interface
x=10
```

2) Write a Java Application to implement final class.

:-

```
final class B{
        void display()
        {
                System.out.println("I am in final class");
        }
}
public class Demo2
{

        public void show()
        {
                System.out.println("I am in show Method");
        }
        public static void main(String[] args) {
        B a1 = new B();
        Demo2 d2 = new Demo2();
   a1.display();
   d2.show();


        }

}
```

Output :

```
I am in final class
I am in show Method
```

3) Write a java application to implement Method Overloading

```
class OverloadDemo {
    // Overload test with no parameters
    void test() {
        System.out.println("No parameters");
    }

    // Overload test for one integer parameter
    void test(int a) {
        System.out.println("a: " + a);
    }

    // Overload test for two integer parameters
    void test(int a, int b) {
        System.out.println("a and b: " + a + " " + b);
    }

    // Overload test for a double parameter
    double test(double a) {
        System.out.println("double a: " + a);
        return a * a;
    }
}
```

```
class Overload {
   public static void main(String args[]) {
      OverloadDemo ob = new OverloadDemo();
      double result;

      // Call all versions of test()
      ob.test();         // no-arg version
      ob.test(10);       // one int argument
      ob.test(10, 20);   // two int arguments
      result = ob.test(123.25); // double argument

      System.out.println("Result of ob.test(123.25): " + result);
   }
}
```

Output:-

```
No parameters
a: 10
a and b: 10 20
double a: 123.25
Result of ob.test(123.25): 15190.5625
```

4) Write a java Application To implement Multithreading.

```
class NewThread extends Thread
{
	public NewThread(String n) {
		super(n);
		start();
	}
	public void run() {
		for(int i=1;i<=5;i++) {
			System.out.println(getName()+"="+i);
			try
			{
				Thread.sleep(500);
			}
			catch(Exception e) {
				System.out.println(e);
			}
		}
	}
}
public class ThreadDemo {
	public static void main(String[] args)
	{
		NewThread n1 = new NewThread("one");
		NewThread n2= new NewThread("two");
		NewThread n3= new NewThread("Three");
		for (int i=1;i<=5;i++)
				System.out.println("Main="+i);
		try {
			Thread.sleep(500);
		}
	catch(Exception e) {
		System.out.println(e);
	}
	}
```

}

Output:-

Main=1

Main=2

Main=3

Main=4

Main=5

One=1

Two=1

Three=1

One=2

Two=2

Three=2

One=3

Two=3

Three=3

One=4

Two=4

Three=4

One=5

Two=5

Three=5

5) Write a java Application to Implement Single level Inheritance

```
class A {
   int x = 10;
   public void display() {
      System.out.println("x = " + x);
   }
}
class B extends A {
   int y = 20;
   public void display1() {
      System.out.println("y = " + y);
   }
}
public class InheritDemo {
   public static void main(String args[]) {
      B b1 = new B();   // Object of subclass B
      b1.display();     // Calls method of class A
      b1.display1();    // Calls method of class B
   }
}
```

Output:-

x=10

y=20

6) Write a Java application to implement passing parameters to an Applet

:-

Applet.html

```
<html>
<body>
<Applet code = "ParamDemo.java" width="400" height="400">
<param name ="string" value ="Alex>
</applet>
</body>
</html>
```

ParamDemo.java

```
import java.awt.*;
import java.applet.*;

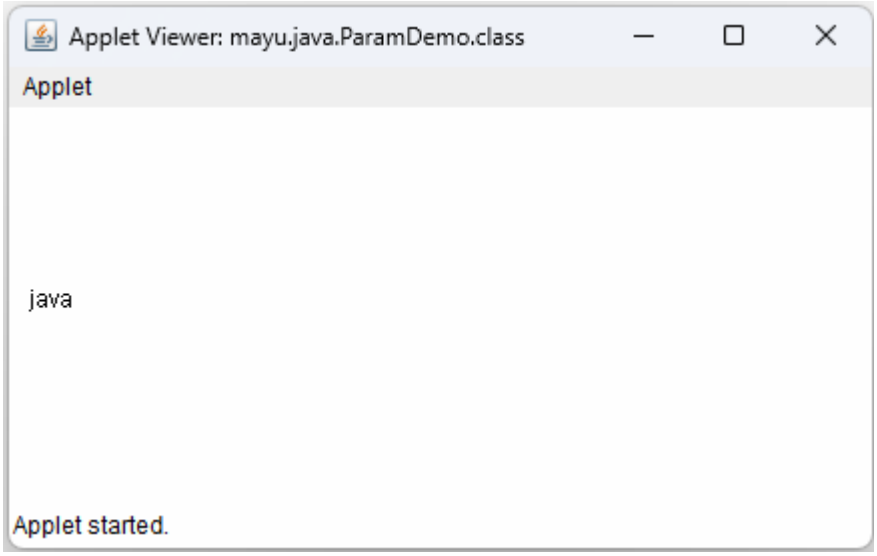
public class ParamDemo extends Applet
{
	String str;
	public void init()
	{
		str=getParameter("string");
		if(str==null)
			str="java";
		else
			str="Hello"+str;
	}

	public void paint(Graphics g)
	{
		g.drawString(str, 10, 100);

	}
```

}

Output :-


Applet Viewer: mayu.java.ParamDemo.class
Applet
java
Applet started.

7) Write an AWT based Java Application for Textarea Control.

```
import java.awt.*;
import java.awt.event.*;

public class TextAreaApp extends Frame {

	TextArea t1, t2;
	Button b1;

	public TextAreaApp() {
		setTitle("my window");
		setSize(400, 400);
		setLayout(null);
		t1 = new TextArea();
		t1.setBounds(20, 50, 200, 150);
		add(t1);

		t2 = new TextArea();
		t2.setBounds(20, 245, 200, 150);
		add(t2);

		b1 = new Button("copy");
		b1.setBounds(20, 210, 70, 25);
		add(b1);

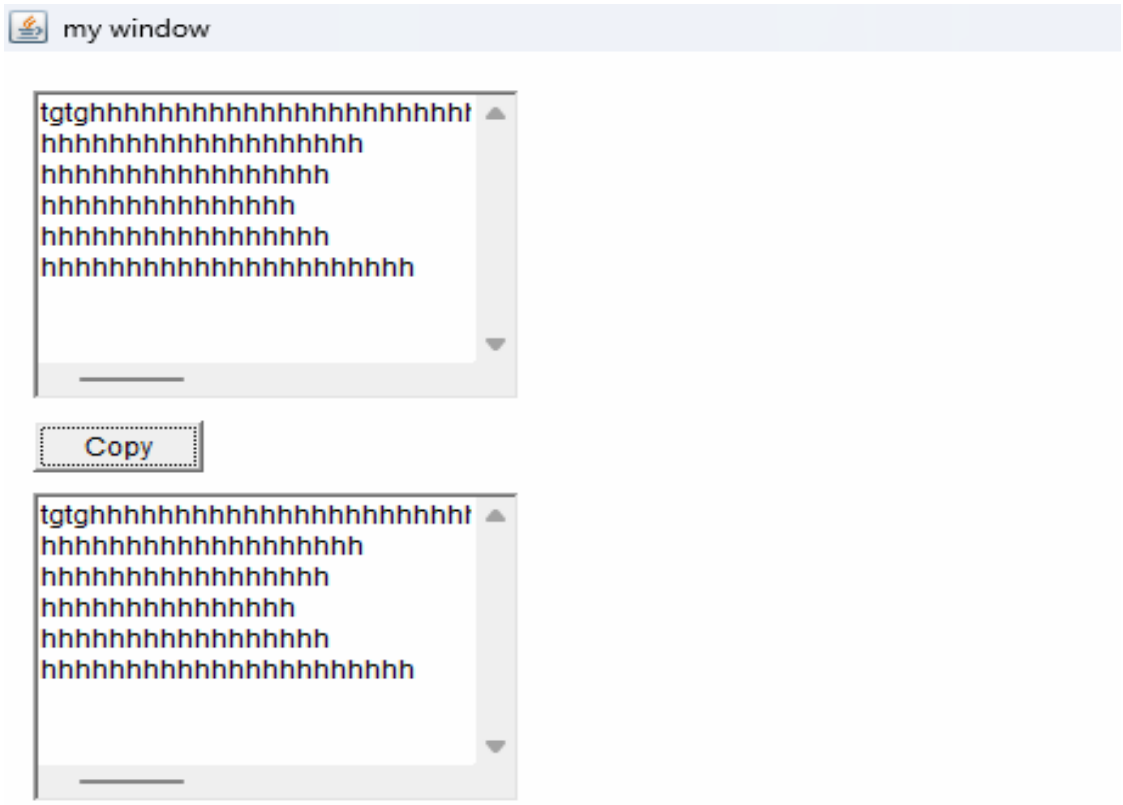
		b1.addActionListener(new ActionListener() {

			public void actionPerformed(ActionEvent ae) {
				t2.setText(t1.getText());
			}
		});
		this.addWindowListener(new WindowAdapter() {
			public void windowClosing(WindowEvent we) {
				System.exit(0);
			}
		});

	}
```

```
        public static void main(String[] args) {
                TextAreaApp f = new TextAreaApp();
                f.setVisible(true);
        }
}
```

Output:-


my window
tgtghhhh

8) Write an AWT based Java application to implement Item Event of Checkbox Control.

```
import java.awt.*;
import
java.awt.event.*;

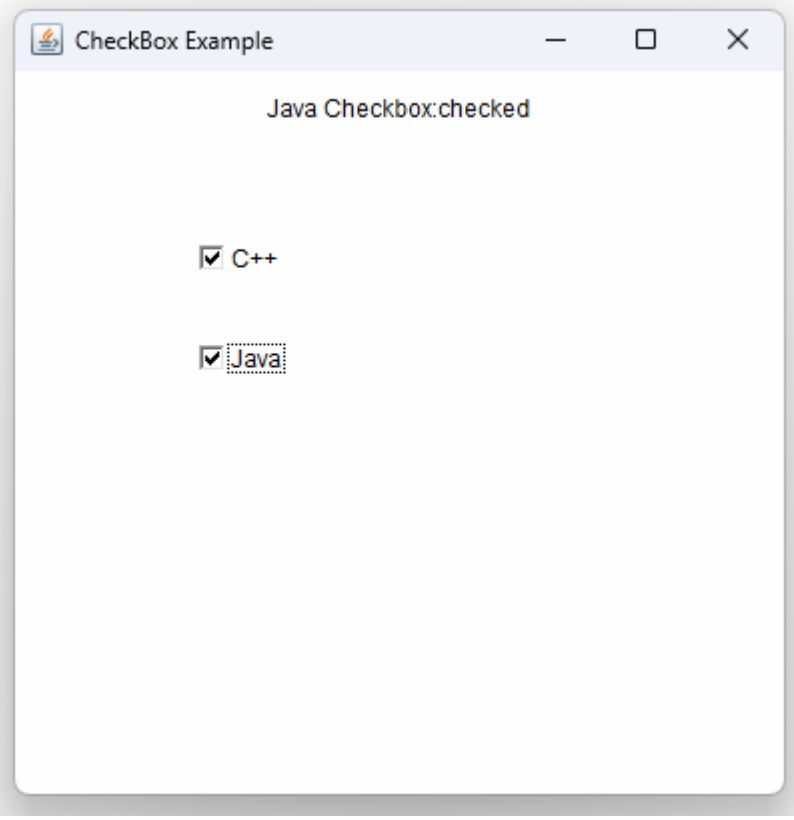
public class MyDemo implements ItemListener
{
	Checkbox checkBox1, checkBox2;
	Label label;
	MyDemo()
	{
		Frame f=new Frame("CheckBox Example");
		label = new Label();
		label.setAlignment(Label.CENTER);
		label.setSize(400,100);
		checkBox1 = new Checkbox("C++");
		checkBox1.setBounds(100,100,50,50);
		checkBox2=new Checkbox("Java");
		checkBox2.setBounds(100,150,50,50);
		f.add(checkBox1);
		f.add(checkBox2);
		f.add(label);
		checkBox1.addItemListener(this);
		checkBox2.addItemListener(this);
		f.setSize(400,400);
		f.setLayout(null);
		f.setVisible(true);
	}
	public void itemStateChanged(ItemEvent e)
	{
		if(e.getSource()==checkBox1)
			label.setText("C++ Checkbox:
"+(e.getStateChange()==1?"checked":"unchecked"));
		if(e.getSource()==checkBox2)
			label.setText("Java
Checkbox:"+(e.getStateChange()==1?"checked":"unchecked"));
```

```
        }

        public static void main(String[] args) {
                new MyDemo();

        }
}
```

Output :-



9) Write a SWING based java application to implement click event of Button Control.

:-

```
import java.awt.event.*;
import javax.swing.*;

public class ButtonExample {
    public static void main(String[] args) {

        JFrame f = new JFrame("Button Example");
        final JTextField tf = new JTextField();
        tf.setBounds(50, 50, 150, 20);

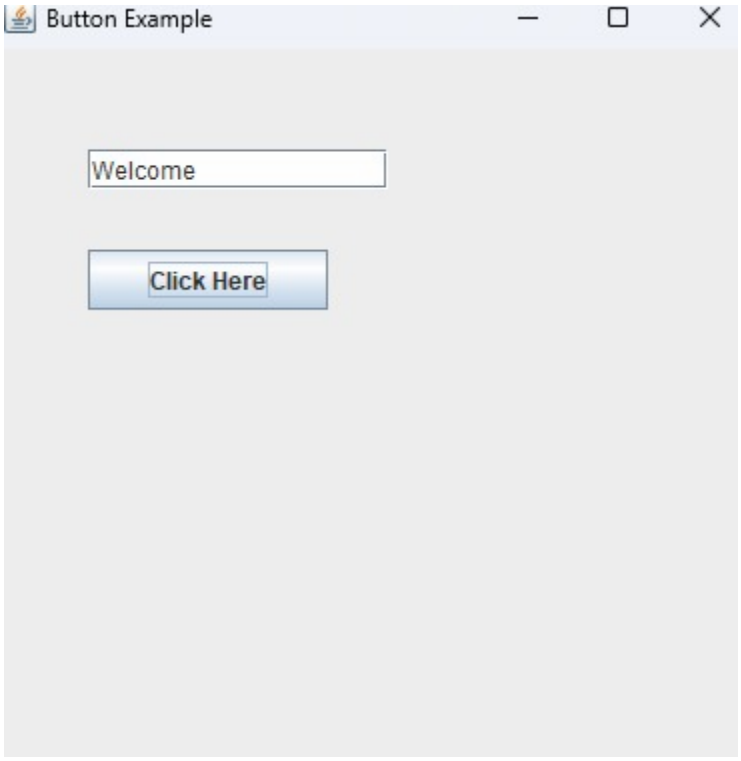
        JButton b = new JButton("Click Here");
        b.setBounds(50, 100, 120, 30);

        b.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                tf.setText("Welcome");
            }
        });

        f.add(b);
        f.add(tf);
        f.setSize(400, 400);
```

```
        f.setLayout(null);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        f.setVisible(true);
    }
}
```

Output:-



10)Write an AWT based Java Application for Menus

```
import java.awt.*;
import java.awt.event.*;

public class AWTMenuDemo {
   private Frame mainFrame;
   private Label headerLabel;
   private Label statusLabel;
   private Panel controlPanel;

   public AWTMenuDemo() {
      prepareGUI();
   }

   public static void main(String[] args) {
      AWTMenuDemo awtMenuDemo = new AWTMenuDemo();
      awtMenuDemo.showMenuDemo();
   }

   private void prepareGUI() {
      mainFrame = new Frame("Java AWT Example");
      mainFrame.setSize(400, 400);
      mainFrame.setLayout(new GridLayout(3, 1));

      mainFrame.addWindowListener(new WindowAdapter() {
         public void windowClosing(WindowEvent windowEvent) {
            System.exit(0);
         }
      });

      headerLabel = new Label();
      headerLabel.setAlignment(Label.CENTER);

      statusLabel = new Label();
      statusLabel.setAlignment(Label.CENTER);
      statusLabel.setSize(350, 100);
```

```
        controlPanel = new Panel();
        controlPanel.setLayout(new FlowLayout());

        mainFrame.add(headerLabel);
        mainFrame.add(controlPanel);
        mainFrame.add(statusLabel);

        mainFrame.setVisible(true);
    }

    private void showMenuDemo() {
        final MenuBar menuBar = new MenuBar();

        Menu fileMenu = new Menu("File");
        Menu editMenu = new Menu("Edit");
        final Menu aboutMenu = new Menu("About");

        MenuItem newMenuItem = new MenuItem("New", new
MenuShortcut(KeyEvent.VK_N));
        newMenuItem.setActionCommand("New");

        MenuItem openMenuItem = new MenuItem("Open");
        openMenuItem.setActionCommand("Open");

        MenuItem saveMenuItem = new MenuItem("Save");
        saveMenuItem.setActionCommand("Save");

        MenuItem exitMenuItem = new MenuItem("Exit");
        exitMenuItem.setActionCommand("Exit");

        MenuItem cutMenuItem = new MenuItem("Cut");
        cutMenuItem.setActionCommand("Cut");

        MenuItem copyMenuItem = new MenuItem("Copy");
        copyMenuItem.setActionCommand("Copy");

        MenuItem pasteMenuItem = new MenuItem("Paste");
        pasteMenuItem.setActionCommand("Paste");

        MenuItemListener menuItemListener = new MenuItemListener();
```

```
        newMenuItem.addActionListener(menuItemListener);
        openMenuItem.addActionListener(menuItemListener);
        saveMenuItem.addActionListener(menuItemListener);
        exitMenuItem.addActionListener(menuItemListener);
        cutMenuItem.addActionListener(menuItemListener);
        copyMenuItem.addActionListener(menuItemListener);
        pasteMenuItem.addActionListener(menuItemListener);

        final CheckboxMenuItem showWindowMenu = new
CheckboxMenuItem("Show About", true);
        showWindowMenu.addItemListener(new ItemListener() {
           public void itemStateChanged(ItemEvent e) {
              if (showWindowMenu.getState()) {
                 menuBar.add(aboutMenu);
              } else {
                 menuBar.remove(aboutMenu);
              }
           }
        });

        // add menu items to menus
        fileMenu.add(newMenuItem);
        fileMenu.add(openMenuItem);
        fileMenu.add(saveMenuItem);

        fileMenu.addSeparator();
        fileMenu.add(showWindowMenu);
        fileMenu.addSeparator();
        fileMenu.add(exitMenuItem);

        editMenu.add(cutMenuItem);
        editMenu.add(copyMenuItem);
        editMenu.add(pasteMenuItem);

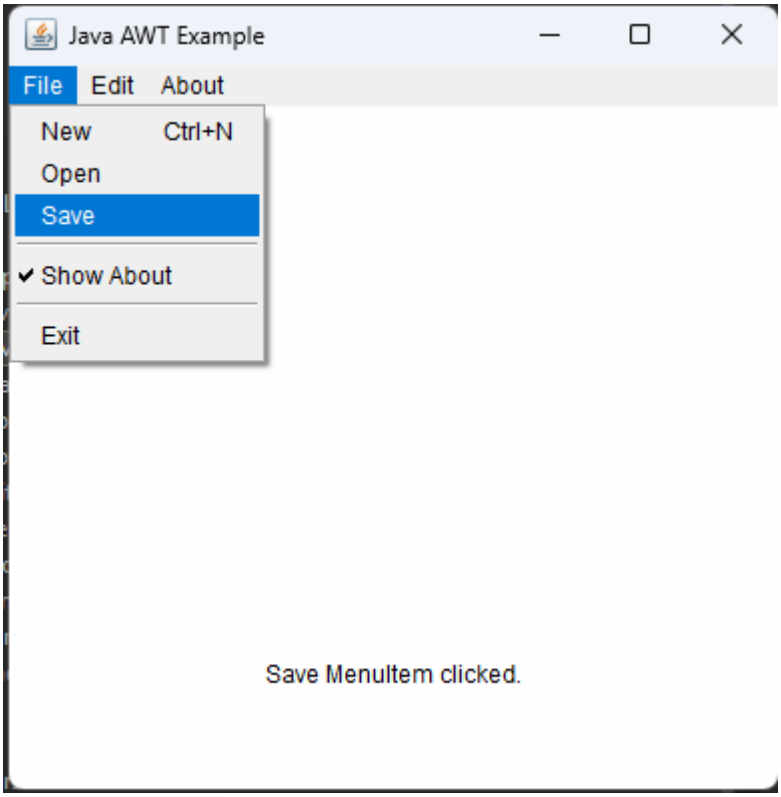
        // add menu to menubar
        menuBar.add(fileMenu);
        menuBar.add(editMenu);
        menuBar.add(aboutMenu);

        // add menubar to the frame
```

```
        mainFrame.setMenuBar(menuBar);
        mainFrame.setVisible(true);
    }

    class MenuItemListener implements ActionListener {
        public void actionPerformed(ActionEvent e) {
            statusLabel.setText(e.getActionCommand() + " MenuItem clicked.");
        }
    }
}
```

Output:-



11)Write a SWING based Java application to implement different Layout Manager.

```
import java.awt.*;
import javax.swing.*;
public class LayoutManagerTest extends JFrame
{
   JPanel flowLayoutPanel1, flowLayoutPanel2, gridLayoutPanel1, gridLayoutPanel2, gridLayoutPanel3;
   JButton one, two, three, four, five, six;
   JLabel bottom, lbl1, lbl2, lbl3;

   public LayoutManagerTest()
   {
      setTitle("Layout Manager Test");
      setLayout(new BorderLayout());

      flowLayoutPanel1 = new JPanel();
      one = new JButton("One");
      two = new JButton("Two");
      three = new JButton("Three");

      flowLayoutPanel1.setLayout(new FlowLayout(FlowLayout.CENTER));
      flowLayoutPanel1.add(one);
      flowLayoutPanel1.add(two);
```

```
flowLayoutPanel1.add(three);



flowLayoutPanel2 = new JPanel();
bottom = new JLabel("This is South");

flowLayoutPanel2.setLayout(new FlowLayout(FlowLayout.CENTER));
flowLayoutPanel2.add(bottom);



gridLayoutPanel1 = new JPanel();
gridLayoutPanel2 = new JPanel();
gridLayoutPanel3 = new JPanel();

lbl1 = new JLabel("One");
lbl2 = new JLabel("Two");
lbl3 = new JLabel("Three");

four = new JButton("Four");
five = new JButton("Five");
six  = new JButton("Six");

gridLayoutPanel2.setLayout(new GridLayout(1, 3, 3, 5));
gridLayoutPanel2.add(lbl1);
gridLayoutPanel2.add(lbl2);
```

```
        gridLayoutPanel2.add(lbl3);

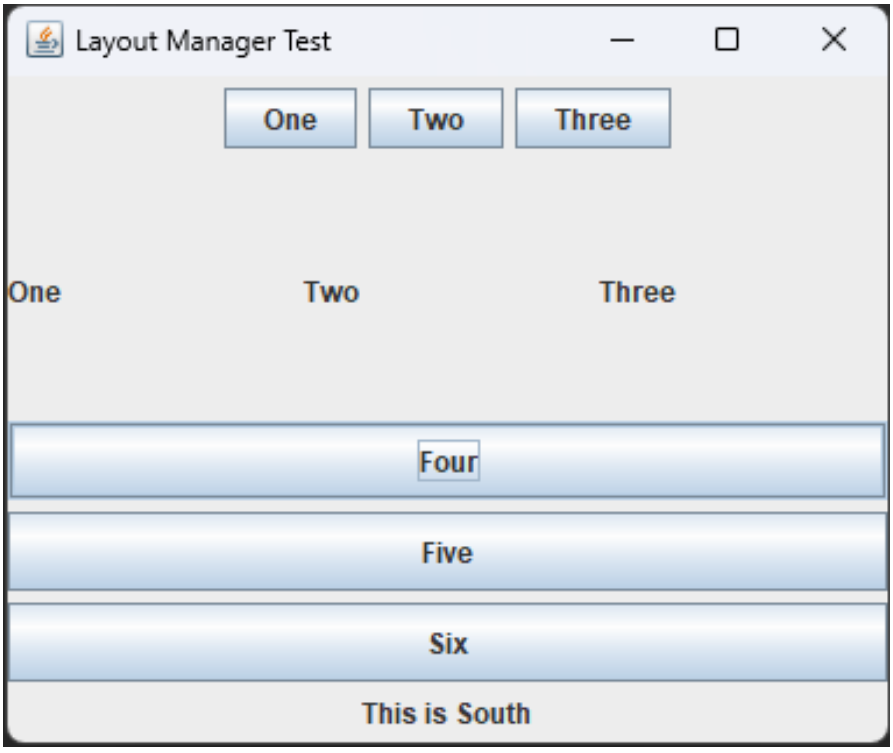
        gridLayoutPanel3.setLayout(new GridLayout(3, 1, 5, 5));
        gridLayoutPanel3.add(four);
        gridLayoutPanel3.add(five);
        gridLayoutPanel3.add(six);

        gridLayoutPanel1.setLayout(new GridLayout(2, 1));
        gridLayoutPanel1.add(gridLayoutPanel2);
        gridLayoutPanel1.add(gridLayoutPanel3);

        add(flowLayoutPanel1, BorderLayout.NORTH);
        add(flowLayoutPanel2, BorderLayout.SOUTH);
        add(gridLayoutPanel1, BorderLayout.CENTER);

        setSize(400, 325);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setVisible(true);
    }
    public static void main(String[] args)
    {
        new LayoutManagerTest();
    }
}
```

Output :-


Layout Manager Test
One
Two
Three
One
Two
Three
Four
Five
Six
This is South

12) Write a java application to create server and client program using TCP/IP protocol.

ServerSide:-

```
import java.net.*;
import java.io.*;
class MyServer
{
public static void main(String args[])
{
try
{
ServerSocket s1=new ServerSocket(420);
Socket s2=s1.accept();
InputStream in=s2.getInputStream();
int ch;
while((ch=in.read())!=-1)
{
System.out.print((char)ch);
}
in.close();
s2.close();
s1.close();
}
catch(Exception e)
{
System.out.println(e);
}
```

```
}

}
```

Output:

```
F:\j-pro>javac MyServer.java

F:\j-pro>java MyServer

good morning mca-msc

F:\j-pro>
```

ClientSide:-

```
import java.net.*;
import java.io.*;
class MyClient1
{
	public static void main(String args[])
	{
		try
		{
			Socket s=new Socket("192.168.6.9",420);
			OutputStream out=s.getOutputStream();
			String str="good morning mca";
			byte b[]= str.getBytes();
			out.write(b);
			out.close();
			s.close();
		}
		catch (Exception e)
		{
			System.out.println(e);
		}
	}

}
```

Output:-

```
good morning mca
```

13) Write a java application to create server and client program using Datagram.

Sender.java

```
import java.net.*;
public class Sender {
   public static void main(String args[]) {
      try {
         DatagramSocket ds = new DatagramSocket(101);
         String str = "good morning";
         byte b[] = str.getBytes();

         DatagramPacket dp = new DatagramPacket(
            b,
            b.length,
            InetAddress.getByName("localhost"),
            102
         );
```

```
            ds.send(dp);
            ds.close();
        }
        catch (Exception e) {
            System.out.println(e);
        }
    }
}
```

Receiver.java

```
import java.net.*;

public class Receiver {
    public static void main(String args[]) {
        try {
            DatagramSocket ds = new DatagramSocket(102);
            byte b[] = new byte[100];

            DatagramPacket dp = new DatagramPacket(b, b.length);
            ds.receive(dp);
            System.out.println(new String(b).trim());

            ds.close();
        }
        catch (Exception e) {
            System.out.println(e);
        }
    }
}
```

Output:-

good morning

14) Write a java application to implement RMI program for addition of two numbers.

→

Adder.java

```
import java.rmi.*;
public interface Adder extends Remote{
        public int add(int x, int y) throws RemoteException;



}
```

AdderRemote.java

```
import java.rmi.*;
import java.rmi.server.*;


public class AdderRemote extends UnicastRemoteObject implements Adder
{
        AdderRemote()throws RemoteException
        {
                super();
        }
        public int add(int x,int y)
        {
```

```
        return x+y;

        }

}
```

MyServer.java

```
import java.rmi.*;
import java.rmi.registry.*;
public class MyServer {

        public static void main(String[] args) {
                try
                {
                        Adder stub = new AdderRemote();
                        Naming.rebind("hello", stub);
                }
                catch(Exception e) {
                        System.out.println(e);
                }

        }

}
```

MyClient.java

```
import java.rmi.*;
public class MyClient {

        public static void main(String[] args) {
        try {
                Adder stub=(Adder)Naming.lookup("hello");
                System.out.println(stub.add(10, 20));
        }
        catch(Exception e) {
                System.out.println(e);
        }

        }

}
```

Output:-

30

15) Write a java application to insert data into database using JDBC.

```
import java.awt.*;
import java.sql.*;
import javax.swing.*;
import java.awt.event.*;

class FormData1 extends JFrame implements ActionListener {
    JLabel l1, l2, l3;
    JTextField t1, t2, t3;
    JButton b1, b2;
    public FormData1() {
        l1 = new JLabel("Enter Name: ");
        l2 = new JLabel("Enter age: ");
        l3 = new JLabel("Address: ");
```

```
t1 = new JTextField(20);
t2 = new JTextField(20);
t3 = new JTextField(20);

b1 = new JButton("Save");
b2 = new JButton("Clear");

b1.addActionListener(this);
b2.addActionListener(this);

JPanel p = new JPanel(new GridLayout(4, 2));
p.add(l1);
p.add(t1);
p.add(l2);
p.add(t2);
p.add(l3);
p.add(t3);
p.add(b1);
p.add(b2);

add(p);
setSize(400, 400);
setVisible(true);
pack();

this.addWindowListener(new WindowAdapter() {
    public void windowClosing(WindowEvent we) {
        System.exit(0);
```

```
        }
    });
}

public static void main(String[] args) {
    new FormData1();
}

public void actionPerformed(ActionEvent e) {

    if (e.getSource() == b1) {
        String v1 = t1.getText();
        int v2 = Integer.parseInt(t2.getText());
        String v3 = t3.getText();

        try {
            Class.forName("sun.jdbc.odbc.JdbcOdbcDriver");
            Connection con = DriverManager.getConnection("jdbc:odbc:mydsn1");

            PreparedStatement ps = con.prepareStatement("INSERT INTO sdata VALUES(?, ?,
?)");
            ps.setString(1, v1);
            ps.setInt(2, v2);
            ps.setString(3, v3);

            ps.executeUpdate();
            JOptionPane.showMessageDialog(null, "Data inserted successfully");
        }
        catch (Exception ex) {
```

```
            System.out.println(ex);
        }
    }
  }
}
```

Output:-

